MIL-L-52325C(ME)
6 December 1977
SUPERSEDING
MIL-L-52325B(ME)
29 June 1971

MILITARY SPECIFICATION

LAUNCHER, TRANSPORTER, BRIDGE: M48A5 or M60A1 TANK CHASSIS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers modifications and equipment to be added to a specially prepared M48A5 or M60A1 tank chassis to make a launcher-transporter for the Armored Vehicle Launched Bridge (AVLB).

1.2 Classification. The bridge launchers shall be the following classification as specified (see 6.2):

M48A5 - Bridge Launcher-Transporter
M60A1 - Bridge Launcher-Transporter

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 5420

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-C-16555	- Coating Compound, Strippable, Sprayable.
MIL-E-19822	- Electrode, Welding, Bare, High Yield Steel.
MIL-E-22200/1	- Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, as Welded or Stress-Relieved Weld Application.
MIL-E-22200/2	- Electrodes, Welding, Covered (Austenitic Chrome-Nickel Steel, for Corrosive and High Temperature Services).
MIL-E-22200/6	- Electrode, Welding, Mineral Covered, Low Hydrogen, Medium and High Tensile Steel.
MIL-E-23765/2	- Electrodes and Rod-Welding, Bare, Solid, Low Alloy Steel.

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MIL-G-23827	- Grease, Aircraft, and Instrument, Gear and Actuator Screw.
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-C-43760	- Cloth, Osnaburg, Cotton, Target, Olive Green 107, Mildew And Water Resistant Treated.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-52464	- Cylinder Assemblies, Actuating, Linear: Hydraulic, for Bridge and Bridge Launcher.
MIL-C-62279	- Chassis, Tank, Armored Vehicle Bridge Launcher, M60A1 and M48A5, Processing for Storage and Shipment of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-882	- System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

DRAWINGS

ME

TA13211E3100	- Launcher, M60A1 Transporting, for Bridge Armored Vehicle Launched.
TA13222E0980	- Launcher, M48A5 Transporting, for Bridge Armored Vehicle Launched.
13211E3255	- Valve Bank, Control.
13222E0979	- Band, Identification.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANS)

ANS B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Description. The M48A5 bridge launcher-transporter, or separately procured component thereof, shall be in accordance with Top Assembly TA13222E0980, and as specified herein; and the M60A1 bridge launcher-transporter, or separately procured components therof, shall be in accordance with Top Assembly TA13211E3100, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the bridge launcher or separately procured components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more M48A5 or M60A1 bridge launchers or separately procured components as specified (see 6.2), for examination testing within the time frame specified (see 6.2), to prove, prior to starting production that his production methods will produce bridge launchers or separately procured components that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor and shall be subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.2.1 Hydraulic system components. In addition to the preproduction model, the following hydraulic components shall be furnished and tested by the contractor prior to the time the preproduction model is furnished: One each of all sequence, pressure relief and check valves, flow regulators, quick disconnect couplings, and one valve bank control. The hydraulic components shall be identical to those furnished on the preproduction model. Tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more M60A1 or M48A5 bridge launchers for inspection as specified in 4.4.

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3.4 Materials. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contracto and shall be subject to all provisions of this specification. Copies of analyses and test reports for all materials shall be available for inspection. Chemical composition and mechanical properties of heat-treated materials shall be verified by chemical analysis and mechanical tests (see 6.10).

3.5 Welding electrodes. Welding electrodes for shielded metal arc and gas shielded arc welding process shall be as specified herein. Electrodes for armor steel, high strength steel, high strength low alloy steel, and structural steel shall be as listed in Table I. When steels of different yield strengths are welded to each other, electrodes specified for the steel having the lowest yield strength shall be used. Electrodes used to weld armor steel or armor steel plate to any other steel shall be electrodes for armor steel or armor steel plate as applicable.

TABLE I. Welding Electrodes

Parent Metal	Welding Process	Specification	Electrode
Armor Steel or High Strength Steel (Yield Strength Not Less than 100,000 psi)	Shielded Metal Arc	MIL-E-22200/1	Type MIL-11018
	Gas Metal Arc	MIL-E-19822 MIL-E-23765/2	MIL-Type B88 MIL-Type 100S-1
High Strength Low Alloy Steel or Structural Steel (Yield Strength Between 36,000-60,000 psi)	Shielded Metal Arc	MIL-E-22200/6	Type MIL-7015 Type MIL-7016
	Gas Metal Arc	MIL-E-22200/1	Type MIL-7018
Armor Steel Plate	Shielded Metal Arc	MIL-E-22200/2	Type MIL-310-15 Type MIL-310-16

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3.6 Hydraulic components.

3.6.1 Cleanliness, and working and proof pressure. Components of the hydraulic system shall be as shown on the applicable drawings and as specified herein. Each component shall be cleaned and flushed prior to testing or installation into the launcher hydraulic system, shall operate at a working pressure of 3,800 psi and shall withstand a proof pressure of not less than 7,000 psi without evidence of permanent deformation or failure.

3.6.2 Cylinder control valves. When tested as specified in 4.6.2.3 each valve bank assembly as shown on Drawing 13211E3255, shall withstand the proof pressure without evidence of external leakage, permanent deformation or failure. After the cylinder control valves have been subjected to the proof pressure, all working sections of the control valve bank assembly shall operate at 3,800 psi without binding or sticking.

3.6.3 Sequencing, pressure relief and check valves. When tested as specified in 4.6.2.4, sequencing, pressure relief, and check relief, and check valves shall withstand the proof pressure without deformation or malfunction. After the components have been subjected to the proof pressure, they shall operate at 3,800 psi within performance limits specified on the applicable drawings. When hydraulic components are procured separately, all openings shall be protected with suitable plastic protectors.

3.6.4 Hydraulic hose. Hydraulic hose shall be as specified on the applicable drawings, and the age of the hose shall not exceed 6 calendar quarters from cure date to date of acceptance by the Government.

3.6.5 Hydraulic cylinders. Hydraulic cylinders shall conform to MIL-C-52464 types, and styles as follows (see 6.4):

Cylinder	Classification
Holddown	Type I
Tongue	Type IV, Style B
Overhead	Type V, Style B
Ejection	Type VI
Locking	Type VII

3.7 Hydraulic system. The hydraulic system of each M48A5 or M60A1 bridge launcher shall operate at a working pressure of 3,800 psi and shall withstand a proof pressure of not less than 7,000 psi without evidence of external leakage (except weepage normally expected of piston

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and packing), permanent deformation, or rupture of any system component. The hydraulic system shall operate using hydraulic oil conforming to MIL-H-6083. The fluid level of the reservoir shall read full when the launcher is in the travel position.

3.7.1 Threaded pipe joints. Apply antiseize polytetrafluoroethylene tape or antiseize compound to all threaded pipe joints. The tape or compound shall be capable of withstanding temperatures between minus 60° F and plus 200° F without deterioration.

3.7.2 System cleanliness. When tested as specified in 4.6.2.5, the presence of any particles on the 100 micron screen shall constitute failure of the test.

3.7.3 Proof pressure test. When tested as specified in 4.6.2.6, the hydraulic system shall withstand the proof pressure of 7,000 psi; nonconformance to 3.7 shall constitute failure of the test.

3.8 Lubrication. The M48A5 and M60A1 bridge launchers shall operate as specified herein when serviced with military lubricants, as specified (see 6.5).

3.8.1 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For ball, roller, and needle bearings, gears and on sliding and rolling surfaces of such equipment as instruments, cameras, electronic gear and mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

(b) MIL-G-10924 - For lubrication and surface corrosion protection of automotive and artillery equipment operated over the temperature range of minus 65° F to plus 225° F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The M48A5 and M60A1 bridge launchers shall be assembled, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

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3.8.2 Oil lubrication. The M48A5 and M60A1 bridge launchers shall operate as specified herein when serviced with military oils designated for use in the temperature range specified herein (see 6.5). A tag shall be attached in a conspicuous place to indicate the temperature range, and the lubricant grade used.

3.9 Identification marking. The M48A5 or M60A1 launcher and separately procured components, as specified, shall be identified in accordance with MIL-STD-130 and as specified on the drawings. All metal die stamping of National Stock Number, nomenclature, and the contract number shall consist of 1/2-inch letters except that small items such as pins that cannot be marked with 1/2-inch letters shall be marked with 3/8-inch or 1/4-inch letters.

3.9.1 Hose assembly and connecting fitting identification. Each hose assembly and each fitting connecting to a hose assembly in the installed hydraulic system shall be provided with identification marking. Each hose assembly shall be provided with a seamless metal marking band, size as shown on Drawing 13222E0979. The seamless marking band shall be die stamped with the following information, letters and numerals shall be not less than 1/8-inch:

Hose assembly code number, and letter.
The size number of the hose.
Date of hose manufacturer.
Drawing number and dash number.

The port openings to which the hose end fittings are connected shall be metal stamped with the code number and letter as specified on the hose assembly drawings. When the hose ends connect to removable fittings such as tees, adapters, and elbows, these connecting fittings shall be fitted with a round brass collar 0.050 inch thick which shall be metal stamped with the appropriate code number and letter using 3/16-inch characters and numerals.

3.9.2 Hose identification. The hose shall be marked with the size number and date of manufacture. The date of manufacture shall include the quarter of the calendar year and the year.

3.10 Treatment and painting. All parts added to the tank chassis (internal and external) shall be treated and painted in accordance with MIL-T-704, Type A, and as specified herein. Masking of the hydraulic hoses is not required during painting of the launcher, and paint on the hoses shall not be cause for rejection.

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(a) The tank chassis exterior shall be painted when damage to the exterior finish occurred during launcher fabrication. All parts and equipment added to the exterior including boom, tongue, cover plates and cupolas (exterior), hydraulic cylinders, bridge seat, radio antenna base and mount, and armor plate shall be prepared and painted in accordance with MIL-T-704, Type A.

(b) After all interior equipment has been installed, all name plates, instruction plates, vision glass, and appropriate portions of moving parts shall be masked with tape and the interior of the hull shall be spray painted with paint conforming to TT-E-529, Color No. 24533. The crew compartment and its cover plates and components may be painted after installation or prior to installation, or may be removed for painting. The engine compartment need not be painted unless the paint finish is damaged during fabrication or installation of the launching mechanism or hydraulic system.

3.11 Government-furnished property. When specified (see (6.2) any or all of the property listed in Table II and when specified (see 6.2) the radio equipment listed in Table III or Table IV as specified shall be furnished by the Government (see 6.6) for installation by the contractor (see 6.6). The contractor shall install the Government-furnished property as shown on the drawings and as specified herein. Government-furnished property not installed shall remain in the shipping containers and shipped attached to the launcher.

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TABLE II. Government-Furnished Property

Item No.	Description	Identification	Quantity	Remarks
1.	M48A5 or M60A1 Bridge Launcher Chassis		See 6.2	
2.	Integral Components Box "A" consisting of the following components	Box "A"		Contractor Installed
	(a) Blower Assembly Ventilator	ORD 7974293 NSN 1015-00-699-9080	1	Contractor Installed
	(b) Box Assembly, Blower Switch	ORD 8335333 NSN 1040-00-675-5071	1	"
	(c) Box, Infra-Red Periscope, Spare Head	ORD 8721352 NSN 2540-00-740-3959	1	"
	(d) Box, Infra-Red Drivers Periscope	ORD 10887134 NSN 2540-00-999-7034	2	"
	(e) Bracket, 5 gal., Water Can	MS53052 NSN 2590-00-473-6331	1	"
	(f) Clamp, Blower	MS21920-78 NSN 4730-00-784-1273	1	"
	(g) Cover, Turret Vent Blower	ORD 7363896 NSN 1015-00-003-9728	1	"
	(h) Filter Unit, Gas Particulate 12 CFM M8A3	MIL-F-51069 NSN 4240-00-853-3201	1	"
	(i) Gasket, Blower	ORD 7364008 NSN 5330-00-736-4008	1	"
	(j) Harness Installation, Radio Set ANVRC-46 & ANVRC-64	NSN 5820-00-494-6639	1	"
	(k) Lead Assy., Turret Vent Blower	ORD 10952142-1 NSN 2590-00-918-6221	1	"
	(l) Lead Assy., Turret Vent Blower	ORD 1095142-2 NSN 2590-00-918-6222	1	"
	(m) Ring Assembly, Blower	ORD 8734283 NSN 1015-00-699-9075	1	"

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TABLE II. Government-Furnished Property (Cont'd)

Item No.	Description	Identification	Quantity	Remarks
	(n) Screw	ORD 7970962 NSN 5305-01-006-2069	4	"
	(o) Silencer Assembly, Blower	ORD 8710143 NSN 2590-00-229-5721	1	Contractor Installed
	(p) Strap, Blower	ORD 7321359 NSN 2590-00-732-1359	1	"
3.	Integral Components	Box "B"	1 set	Boxed separately and not opened
4.	Sling, Multiple Leg	NSN 5420-00-296-9165	2	Contractor Placed
5.	Basic Issue Items	M48A5 or M60A1 as applicable	1 set	Boxed separately and not opened
6.	Expendable Supplies and Materials	"	1 set	"
7.	Additional Authorized Equipment	"	1 set	"

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TABLE III. Government-Furnished Radio Property (AN/GRC)

Item No.	Description	Identification	Quantity for each launcher
1.	Basic unit radio set AN/GRC-3	5820-00-543-1784	1 set
	or -4	5820-00-243-6418	
	or -5	5820-00-230-0449	
	or -6	5820-00-222-6417	
	or -7	5820-00-230-0446	
	or -8	5820-00-230-0454	
2.	Installation unit for radio sets AN/GRC-3 through -8 in M60A1 launcher consisting of:		
	(a) Accessory assembly	Scintilla Magneto Div., Bendix Aviation Corp., No. 10-229936-225 5975-00-823-0897	1
	(b) Adapter UG-306B/U	5935-00-171-3016, MS35368	2
	(c) Cable assembly kit (for terminating cable, special purpose) WM-60/U	5995-00-356-0179	3
	(d) Cable assembly (24 ft. 0 in. long)	SC-C-75348	1
	(e) Cable, special purpose, WM-46/U	MIL-S-2486 6145-00-184-3107	15 ft.
	(f) Chest set group AN/GSA-6	5965-00-329-9005	3
	(g) Clamp strap kit	5820-00-752-5738	3
	(h) Control box, C-375/VRC, PPL#1537	MIL-C-12884	2
	(i) Headset-microphone H-63/U	MIL-H-11190	3
	(j) Plug assembly	Scintilla Magneto Div., Bendix Aviation Corp., No. 10-229935-2P	1
	(k) RF cable assembly CG-530D/U (17 ft. long)	SC-C-23604	2
	(l) Rope RP-5	T-C-571	25 ft.

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TABLE IV. Government-Furnished Radio Property (AN/VRC)

Item No.	Description	Identification	Quantity for each launcher
1.	Basic unit radio set AN/VRC-46	5820-00-892-0871	1
2.	Installation unit for radio set AN/VRC-46 in M60A1 launcher, consisting of:		
	(a) Accessory assembly	Scintilla Magneto Div., Bendix Aviation Corp., No. 10-229936-225 5975-00-823-0897	1
	(b) Angle bracket assembly	SC-C-48605	2
	(c) Cable assembly, radio-frequency CG-1773/U (8 ft lg)	SM-D-415563 5995-00-823-2990	1
	(d) Cable assembly, S. P. Electrical CX-4722/VRC (8 ft lg)	SM-D-415552 5995-00-823-2819	1
	(e) Clamp strap kit, PPL No. 914	5820-00-752-5738	1
	(f) Connector (waterproof) assembly	SC-C-28203 (Group I)	1
	(g) Connector (waterproof) assembly	SC-C-49627	1
	(h) Plug assembly	Scintilla Magneto Div., Bendix Aviation Corp., No. 10-229935-2P 5935-00-856-8426	1
3.	Basic unit intercommunications set AN/VIC-1(V)	5830-00-856-3273	1
4.	Installation unit for AN/VIC-1(V) in M60A1 launcher consisting of:		
	(a) Bracket assembly	SC-D-48664 5820-00-875-0920	1
	(b) Cable assembly, S. P. Electrical CX-4723/VRC (5 ft lg)	SM-D-415553 5995-00-823-2833	2
	(c) Cable assembly, S. P. Electrical CX-4723/VRC (6 ft lg)	SM-D-415553 5995-00-823-2834	1

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TABLE IV. Government-Furnished Radio Property (AN/VRC) (Cont'd)

Item No.	Description	Identification	Quantity for each launcher
(d)	Clamp strap kit, PPL No. 914	5820-00-752-5738	3
(e)	Extension audio cable assembly (lower cord assembly for H-161/GR) 40 ft lg	MIL-C-5517	1
(f)	Headset-microphone H-161/GR	MIL-E-55119 5965-00-876-2376	3

3.12 Government-loaned property. Unless otherwise specified, (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.7):

ITEM NO.	DESCRIPTION	IDENTIFICATION	QUANTITY
1	63-foot Armored Vehicle Launched Bridge	NSN 5420-00-522-9599	1
2	Inspection Equipment for Launcher-Transporter	Quality Assurance Pamphlet P715-104	1 set
3	QA pamphlet Care Operation and Maintenance of Inspection Equipment for Launcher M48A5/M60A1 Tank Chassis Transporting:	P715-104	1

3.13 M48A5 Bridge launcher-transporter. The M48A5 bridge launcher transporter shall conform to Top Assembly TA13222E0980. When tested as specified in 4.6.2.7, 4.6.2.8, or 4.6.2.9 as applicable, each M48A5 bridge launcher shall transport, launch, disconnect, and recover the armored vehicle launched bridge (AVLB). Nonconformance to 4.6.2.7, 4.6.2.8, or 4.6.2.9 shall constitute failure of the test.

3.14 M60A1 bridge launcher-transporter. The M60A1 bridge launcher transporter shall conform to Top Assembly TA13211E3100. When tested as specified in 4.6.2.7, 4.6.2.8, or 4.6.2.9 as applicable, each M60A1 bridge

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launcher shall transport, launch, disconnect, and recover the armored vehicle launched bridge (AVLB). Nonconformance to 4.6.2.7, 4.6.2.8, or 4.6.2.9 shall constitute failure of the test.

3.15 Reliability. The Mean-Time-Between-Failures (MTBF) shall be not less than 138 launch and retrieve cycles when tested as specified in 4.6.2.8. A cycle shall consist of one launch and one retrieval of the Armored Vehicle Launched Bridge (AVLB). When the M48A5 or M60A1 bridge launcher is tested as specified in 4.6.2.9 the launcher will be acceptable if not more than three defects are present. The sample size shall be as specified (see 3.3 and 4.4).

3.16 Maintenance ratio. The ratio of man hours of maintenance required to the hours of operation performed shall be not greater than 0.29.

3.17 Safety and human factors.

3.17.1 Safety. Within the limits of prescribed design specifications, operation and maintenance of the M48A5 or M60A1 bridge launcher shall not create hazard levels greater than those defined in MIL-STD-882.

3.17.2 Human factors. Within prescribed design specifications, human factors considerations shall be in accordance with MIL-STD-1472.

3.18 Workmanship.

3.18.1 Steel fabrication. Steel used in the fabrication of the launcher shall be free from laminations, kinks, and sharp bends. Corners shall be square and true. All burrs and rough edges shall be removed. All bends shall be made to a radius which will not cause cracking at the extreme outer fibers of the piece being bent, and all bends except for minor details shall be made by the use of metal dies or fixtures to insure uniformity of size and shape. Shearing, chipping, and flame cutting shall be done neatly and accurately. Prior to flame cutting of high strength or armor steels, the contractor shall prepare a flame-cutting procedure. This procedure, submitted in duplicate, shall be subject to the approval of the contracting officer.

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3.18.2 Machined surfaces. Machined surfaces shall conform to ANS B46.1.

3.18.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have all burrs and rough edges removed.

3.18.4 Welders, welding, and care of welding electrodes.

3.18.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification, Welding Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

3.18.4.2 Welding. Welding of high-strength, low-alloy, and armor steels shall be in accordance with MIL-STD-1261, Class 3. Welding of structural steels shall be in accordance with MIL-STD-1261, Class 1. The workmanship specimens required by MIL-STD-1261, Class 3 welds shall be samples simulating all welding conditions and welding actual materials to be welded.

3.18.4.3 Storage and handling of welding electrodes. The welding procedures required by MIL-STD-1261 shall also include a description of the safeguards used to store and handle welding electrodes. The contractor shall provide a detailed description of the safeguards used for bare wire electrodes to be used with the gas metal arc process. Coated stick electrodes shall be stored and handled as follows:

(a) When a container of welding electrodes is opened, the entire contents shall be placed immediately in a baking oven, with electrodes placed not more than three layers deep on a tray, and baked either (1) for a minimum of 1 hour, but not more then 1-1/4 hours at 800° F; or (2) for a minimum of 4 hours at 400° F.
(b) The baked electrodes, while still warm, shall then be placed in a ventilated holding oven set at 250° to 300° F.
(c) Not more than a 1/2-hour supply of welding electrodes shall be taken from the holding oven at one time.
(d) If electrodes are out of the holding oven in excess of 1/2 hour but less than 1 hour, they must be returned to the holding oven for a period of not less than 1 hour.

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(e) If electrodes are out of the holding oven in excess of 1 hour, they shall be returned to the baking oven and rebaked as described under (a).
(f) Electrodes removed for use shall not be allowed to come in contact with any moist or wet surfaces. Electrodes that have been in contact with water, grease, or oil shall be scrapped.

3.18.4.4 Welding fixtures. Shop fabricated components (except for minor parts) shall be assembled in steel fixtures or frames and welded while held securely in position. Minor parts may be held together by any means that will insure secure and proper positioning. The fixtures or frames shall be designed to minimize distortion of the parts being welded. Subsequent straightening and stress relieving of components fabricated from log-alloy and structural steels will be permitted if the straightening process does not damage the end item. Subsequent straightening of components fabricated from high-strength steel will be permitted; however, stress relieving shall not be permitted.

3.18.5 Electrical wiring and mechanical linkages. Any existing instruments, gages, panels, lights, mechanical linkages, and accessory items which must be removed for installation of the launcher equipment shall be reinstalled in complete operating condition.

3.18.6 Interchangeability. All like commercial components furnished by the contractor shall be directly interchangeable with respect to installation, performance, and quality. The fabricated assemblies shall be interchangeable between launchers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings as applicable.

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4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-furnished and Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.8 and 6.9).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction M48A5, M60A1 bridge launcher or separately procured component as specified shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction M48A5, M60A1 bridge launcher or separately procured component as specified shall be subjected to the tests marked "X" in Column 2 of Table V. Failure of any test shall be cause for performing the inspection in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production M48A5 or M60A1 bridge launchers as specified will be selected at random from the M48A5 or M60A1 bridge launchers being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 3 of Table V

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to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production bridge launcher shall not exclude the remaining bridge launchers from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all tests contained in this specification as part of the initial production test, and failure of additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production M48A5 or M60A1 bridge launcher as specified, to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production bridge launcher and shall be cause for refusal by the Government to continue acceptance of production bridge launchers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on bridge launchers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all bridge launchers accepted prior to completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination of bridge launcher components and for examination and test of hydraulic components and launcher components shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Individual. Each M48A5 or M60A1 bridge launcher or separately procured component shall be examined for the applicable defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2.2 Samples. Samples of separately procured components and hydraulic and launcher components selected in accordance with 4.4.1 shall be examined for the applicable defects specified in 4.6.1. AQL shall be 2.5 percent defective.

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4.5.3 Tests.

4.5.3.1 Individual. Each M48A5 or M60A1 bridge launcher or separately procured component shall be subjected to the applicable tests marked "X" in Column 4 of Table V. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples of separately procured component and hydraulic component shall be subjected to the applicable tests marked "X" in Column 4 of Table V. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Each M48A5 or M60A1 bridge launcher or separately procured components as specified shall be examined as specified herein for the following defects as applicable:

101. Any part or component not in accordance with the SQAP requirements as shown on the drawing.
102. Any dimension other than those identified on the SQAP requirements shown on the drawings not as specified.
103. Materials not as specified and analysis and tests missing.
104. Welding electrodes not as specified.
105. Hydraulic components not cleaned and flushed as specified.
106. Age of hydraulic hose not as specified.
107. Hydraulic cylinders not as specified.
108. Hydraulic system unable to operate at the working pressure as specified.
109. Hydraulic system unable to withstand the proof pressure as specified.
110. Antiseize tape or compound not applied to pipe threads as specified.
111. Lubricants not as specified.
112. Identification marking not as specified.
113. Hose assembly and connecting fitting identification marking not as specified.
114. Hose age identification not as specified.
115. Treatment and painting not as specified.
116. Government-furnished property not installed as specified.
117. Radio equipment not installed as specified, when radio installation is specified.
118. Workmanship not as specified.

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4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed at an ambient air temperature between 40° F and 90° F except that the launcher performance test shall be performed without shelter at climatic conditions existing at the test location. The M48A5 or M60A1 bridge launcher shall operate as specified without maintenance other than normal scheduled maintenance established by the applicable technical manuals (see 6.11). Unless otherwise specified herein all testing shall be accomplished using hydraulic oil conforming MIL-H-6083.

4.6.2.2 Test schedule.

TABLE V. Test Schedule

Separately Procured Components	Preproduction Model	Initial Production	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	-	X	Cylinder control valves.	4.6.2.3	3.6.2
X	X	-	X	Hydraulic component proof pressure test.	4.6.2.4	3.6.3
-	-	-	X	Hydraulic system cleanliness.	4.6.2.6	3.7.2
-	-	-	X	Hydraulic system proof pressure.	4.6.2.7	3.7.3
-	X	-	-	Preproduction performance.	4.6.2.8	3.13, 3.14
-	-	X	-	Initial production performance.	4.6.2.9	3.13, 3.14
-	-	-	X	Production performance.	4.6.2.10	3.13, 3.14

4.6.2.3 Cylinder control valves proof pressure test.

4.6.2.3.1 Test procedure.

(a) Assemble the valve bank as shown on Drawing 13211E3255.
(b) Subject the valve bank to a static proof pressure of not less than 7,000 psi with the valves in each of the three positions up, down, and neutral.

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(c) Maintain the pressure for not less than 5 minutes for each of the three valve positions.
(d) Lower the pressure to 3,800 psi and elevate the temperature of the oil to 160° F plus or minus 5° F and operate each valve in all three positions.

4.6.2.3.2 Failure criteria. Nonconformance to 3.6.2 shall constitute failure of the test.

4.6.2.4 Hydraulic component proof pressure test.

4.6.2.4.1 Test procedure.

(a) Subject each sequence, pressure-relief, and check valve, quick disconnect coupling and flow regulator to a proof pressure of not less than 7,000 psi for not less than 5 minutes. Quick disconnect couplings must be tested assembled.
(b) Lower the applied pressure and adjust the sequence, pressure-relief, or check valve to the operating pressure shown on the drawings.
(c) Elevate the temperature of the oil to 160° F plus or minus 5° F and subject the valves and flow regulators to the working pressure of 3,800 psi.
(d) Record the actual operating pressure and flow through each of the items listed above under the conditions imposed by (c) above.

4.6.2.4.2 Failure criteria. Nonconformance to 3.6.3 shall constitute failure of the test.

4.6.2.5 Hydraulic system cleanliness.

4.6.2.5.1 Test procedure.

(a) Prior to installing any hydraulic component into the hydraulic system, clean and flush all hydraulic components, as specified in 3.6.1.
(b) Assemble the launcher hydraulic system and assemble bypass lines which will remove the cylinders, relief valves and sequence valves pump and filter from the hydraulic circuit. Connect the quick-disconnect fittings at the end of the tongue assembly with a short hose assembly.
(c) Connect the hydraulic circuit described above to an external hydraulic system through the slave connections to an external hydraulic system that has a system capacity of 30 gpm and capable of 7,000 psi for a duration of not less than 5 minutes.

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(d) Insert a removable 100 micron filter screen in the return line leading from the launcher to the external reservoir. No other filters shall be installed in the circuit. Activate the pump and develop a flow of 30 gpm. Hold each of the cylinder control valves in the raised and lowered position for not less than 5 minutes in each position.
(e) Disassemble the filter and examine the filter screen for particles.
(f) If particles are present on the 100 micron filter screen, continue flushing the system until conformance to 3.7.3 is achieved.

4.6.2.5.2 Failure criteria. Nonconformance to 3.7.3 shall constitute failure of the test.

4.6.2.6 Hydraulic system proof pressure.

4.6.2.6.1 Test procedure.

(a) After the hydraulic system has been flushed and cleanliness conforms to 3.7.3, reconnect all cylinders into the hydraulic system. Remove the clevis pin from the rod end of the tongue and overhead cylinder and allow the pistons to seat against the cylinder head. The rod will be completely extended, unrestrained and free to more through its entire length.
(b) Cycle each cylinder control valve to the up and down position not less than five times.
(c) Apply and slowly increase the pressure to not less than 7,000 psi and maintain the pressure for not less than 5 minutes.

4.6.2.6.2 Failure criteria. Nonconformance to 3.7 and 3.7.4 shall constitute failure of the test.

4.6.2.7 Preproduction performance.

4.6.2.7.1 Test procedure.

(a) Adjust each pressure relief valve to the operating pressure specified on the drawings using the launcher pump to apply pressure. The pump shall be driven at an engine speed of 1,500 rpm when the valves are adjusted.
(b) Connect the launcher to the bridge and perform a total of 200 launching and retrieval cycles under the following conditions:

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(1) One-hundred launchings and recoveries with launcher on level ground. During this test, the ejection cylinders shall be used to disconnect the bridge for five of the launchings when the bridge is unfolded. Conduct 50 percent of these launchings and recoveries with the launcher at each end of the 63-foot bridge.

(2) Sixty launchings and recoveries with the launcher on a slope: 30 at an upward slope of 15 percent and 30 at a downward slope of 15 percent. While launching on the downward slope, an attempt shall be made to eject the bridge by operating the ejection cylinders with the lock cylinder plugs extended. Conduct 50 percent of the launchings and recoveries with the launcher at each end of the 63-foot bridge.

(3) Forty launchings and recoveries with launcher on a transverse side slope of 8 percent. Conduct 50 percent of the launchings and recoveries with the launcher at each end of the 63-foot bridge. During one of the above launchings, the launcher shall be positioned so that the piston of the tongue cylinder is seated against the cylinder head. Apply pressure until the relief valve operates.

(c) After completion of the launch and retrieve cycles, repeat the procedure stated in (a) above and check and adjust the pressure relief valves, if necessary, to the operating pressure specified on the applicable drawings.

4.6.2.7.2 Failure criteria. The M48A5 or M60A1 bridge launcher shall launch and retrieve the bridge as specified in 3.13 or 3.14 as applicable without evidence of the following defects:

119. Malfunction of components, such as, but not limited to binding or sticking of parts or improper operation.
120. Permanent deformation of any component.
121. Cracks in welds or base metal.
122. Leaks at any point in the hydraulic system.
123. Binding or sticking of any mechanical or automotive linkages.
124. Improper sequence of operation of the ejection cylinders.
125. Interference between the bridge and bridge beam seat.

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4.6.2.8 Initial production performance.

4.6.2.8.1 Reliability.

4.6.2.8.1.1 Test procedure. Follow the test procedure outlined in 4.6.2.7.1 except that the 200 launch and retrieval cycles shall be interspersed throughout the test on a basis of 50 km for each 10 cycles. The cycles shall be made on level ground and on the slopes as outlined in 4.6.2.7.1.6. The launcher shall transport the bridge a total of 1000 kilometers (km) over paved and secondary roads and cross country as follows:

Kilometers	Terrain	Launch/Retrieve Cycles	Max speed km/hr
400	Paved road	80	55
500	Secondary road	100	55
100	Cross country	20	25

After completion of the above transport and launch and retrieve cycles, check and, if necessary, adjust the pressure relief valves to the operating pressure shown on the applicable drawings.

4.6.2.8.1.2 Failure criteria. When the launcher is tested in accordance with the above procedure the launcher undergoing test shall be acceptable if three or fewer failures are present. For the purpose of this test a failure is defined as any malfunction that causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance capability of the launching mechanism or hydraulic system below mission levels.
(b) Serious damage to the launching mechanism by continued operation.
(c) Serious personnel safety hazard.

Damage to the launching mechanism or hydraulic system caused by accidents, improper operation or improper maintenance shall not be considered as failures. Damage due to these causes shall be repaired and the test continued.

4.6.2.8.2 Maintenance evaluation. The ratio of all man hours of maintenance required to the hours of operation performed during initial production reliability testing shall be computed. Nonconformance to 3.16 shall constitute failure of the test.

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4.6.2.9 Production performance.

4.6.2.9.1 Test procedure.

(a) Prior to performing the test adjust each pressure relief valve as specified in 4.6.2.7.1(a).
(b) Connect the launcher to the end of the bridge opposite the end containing the scissoring cylinder and launch and retrieve the bridge not less than 20 times on level ground.
(c) Disconnect the bridge from the launcher using the ejection cylinder five of the 20 launchings.
(d) During one of the 20 launchings, position the launcher so that the piston of the tongue cylinder is seated against the cylinder head. Apply pressure until the relief valve operates.

4.6.2.9.2 Failure criteria. Evidence of any of the defects specified in 4.6.2.7.2 shall constitute failure of the test.

4.7 Inspection comparison. The Government may select M48A5 or M60A1 launchers or separately procured components at any time during the contract production period and subject the M48A5 or M60A1 bridge lanuchers or separately procured components to the examination specified in 4.6.1 and the tests marked "X" in Column 3 of Table V to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on M48A5 or M60A1 or separately procured components selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production bridge launchers or separately procured components. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests as specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison of M48A5 or M60A1 bridge launcher or separately procured components to meet any requirement specified herein during and as a result of examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison bridge launcher or separately procured components and shall be cause for refusal by the Government to continue acceptance of production launchers or separately procured components until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be

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accomplished by the contractor at no cost to the Government on bridge launchers or separately procured components previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all bridge launchers or components accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Procurement of complete launchers. Launcher components shall be subjected to inspection and tests specified in the quality assurance provisions section of MIL-C-62279. Components peculiar to the launcher for which no specific inspection is specified in MIL-C-62279 shall be inspected in accordance with the quality assurance provisions established in MIL-C-62279 and the technical requirements specified herein.

4.8.2 Procurement of launcher components. When only launcher components are procured; the preservation, packing, and marking shall be examined to determine compliance with the Section 5 requirement. Presence of one or more defects shall be cause for rejection.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Procurement of complete launchers. Each launcher shall be preserved in accordance with the applicable Level A requirements of MIL-C-62279, except a vehicle protective closure and crew compartment cover shall not be required, and in accordance with the following additional requirements.

5.1.1.1.1 Hydraulic system. The pistons shall be retracted as far as possible. Exposed portions of the piston rods and the steel thread protectors (if cylinders are shipped without clevis) shall be coated with grease conforming to MIL-G-10924, and the coated portions shall be wrapped or covered with barrier material conforming to MIL-B-121, Grade A, Type I, Class 2, extending the wrap approximately 2 inches on the cylinder head. The wrap shall be secured in place with tape conforming to PPP-T-60, Type IV. Exposed surfaces of pins (plugs) and pin holes (plug

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holes) shall be painted in accordance with 3.10, or coated with preservative conforming to MIL-C-16173, Grade 1. Open ends of hydraulic hose shall be sealed with tape specified herein or plastic caps conforming to MIL-C-5501, Type I. Dust caps shall be installed on open ends of hydraulic hose. (Note: When piston rods are fully retracted, there will be no exposed surfaces.)

5.1.1.1.2 Drain plugs. Drain plugs in the four fender toolboxes shall be removed and taped to inside of box with tape conforming to PPP-T-60, Type IV.

5.1.1.1.3 Tools. Tools shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.1.4 Periscope assembly opening. The entire perimeter of the rectangular cover as well as the area over the hinge shall be covered with tape conforming to PPP-T-60, Type IV.

5.1.1.1.5 Cupola cover opening. The cupola door shall be shut and tightly secured by adjustment of the latch. After being tightly closed and secured, the joint between the cupola cover and vision block ring shall be sealed with tape conforming to PPP-T-60, Type IV.

5.1.1.1.6 Antenna receptacle connectors. The antenna receptacle connectors (RF and system cable receptacle connectors J 301 and J 310) shall be preserved by covering with a waterproof bag or pouch of suitable size which shall be taped to the cables with tape conforming to PPP-T-60, Type III. A waterproof plastic cushioning shall be secured around the cables before bag is taped in place. Note: Remove armor cover to have access to cables.

5.1.1.1.7 Engine and exhaust grills. The engine and exhaust grills shall be given a protective cover consisting of osnaburg cloth conforming to MIL-C-43760 sprayed with a strippable compound conforming to MIL-C-16555, Type II, Class 1.

5.1.1.1.8 Miscellaneous supplies and equipment. Items, equipment, and components not specifically identified herein which require protection from corrosion and mechanical damage shall be preserved as required for items, equipment and components of similar design and construction or preserved in accordance with the preservative application criteria and applicable method(s) of preservation of MIL-P-116.

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5.1.1.1.9 Consolidation.

5.1.1.1.9.1 M60A1. Preserved and unpreserved items shall be consolidated to the degree necessary to enable items located within the vehicle to be placed to the maximum extent within the three personnel baskets. It is necessary, in the left and right side baskets, to use a dunnage strip approximately 3 inches in height to counteract the slope of the bottom and to allow the containers placed within the baskets to have a firm bottom rest (with aid of the dunnage). Certain large sized tool items shall be placed on the floor next to the assistant's seat and secured to the bracket to the left of the seat. The maximum possible size for a consolidation pack for the right side basket is: 20-1/4 inches, length, 10-1/2 inches, width, and 16-1/2 inches height, all being outside dimensions. When the periscope is placed in a container with maximum I.D. dimensions of: 19-1/2 inches, length X 9-1/2 inches width X 12-1/4 inches depth, one can be located in the left side basket and the other, along with the two HEAD pieces, can be located in the front center basket. All items located in the three baskets should be secured therein by tieing a suitable cotton cord or tape over the top of the container in the basket and securing the cord to each side of the perforated baskets. The filter unit, gas particulate, which will not be installed, but shipped with the unit, shall be located and secured in any available space within the vehicle. The OVM box (approximately 38 inches in length x 22 inches in width x 17 inches in depth, inside dimensions) shall conform to requirements of PPP-B-621, Style 2 for Types 1 and 2 loads or PPP-B-601, Overseas Type, Style I, grade optional (net load of approximately 500 pounds). Items to be packed in this OVM box are as follows: Cable assembly, towing, 2; can, water; flag staffs, 3; hook, tow cable, 3; pin, tow cable, hook, 3; ax, single bit; sledge hammer; mattock handle and pick; fixture, track conn., 2; puller assembly, track link and connector; wrench (5120-00-563-7342); sling, multiple leg, 2; wrench (5920-00-542-3144); wrench (5420-00-050-8745); tow hook and pin, 2; hydraulic slave hose. Three girthwise straps 3/4 inch X 0.023 inch, minimum, conforming to requirements of QQ-S-781, Class 1, Type I or IV, Finish B shall be used. The box shall be located to the rear of the bridge beam seat on the engine cover, but abutting the bridge beam seat. Dunnage of suitable thickness shall be used between the box and the bridge beam seat to compensate for the protrusion of the holddown cylinder pin. Box shall be secured with two lengthwise straps, running from the top deck lifting rings located at both ends of the box over the top of the box and sealed in the center of the box. In addition, a girthwise strap shall be placed around the circumference of the box and the bridge guides (which shall be protected with strapping edge protectors) and then sealed on the rear side of the box. Strapping shall conform to requirement above.

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5.1.1.1.9.2 M48A5. Consolidation shall be as specified in 5.1.1.8.1 for the M60A1 to the maximum extent possible. Because of only two personnel baskets and limited stowage space, it is realized that the requirements cannot be followed exactly. Components that cannot be consolidated as specified in 5.1.1.8.1 shall be placed and secured in any available space that will not interfere with lifting, towing, or steering.

5.1.1.2 Procurement of launcher components. When only launcher components are procured, they shall be preserved in accordance with the preservative application criteria and the most applicable method as specified in MIL-P-116.

5.1.2 Level C.

5.1.2.1 Procurement of complete launchers. Each launcher shall be preserved in accordance with Level B requirements of MIL-C-62279, except a crew compartment cover shall not be required; the engine and exhaust cover shall be fabricated from waterproof paper or plastic film, secured in place with waterproof tape; and in accordance with the additional requirements specified herein for Level A.

5.1.2.2 Procurement of launcher components. When only launcher components are procured they shall be preserved in a manner to afford protection against deterioration and mechanical and physical damage from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Procurement of complete launchers. The launchers shall be shipped without additional packing and shall be loaded on open-top rail cars in accordance with the loading requirements of MIL-C-62279.

5.2.1.2 Procurement of launcher components. When only launcher components are procured, they shall be preserved as specified in 5.1 and packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional; PPP-B-621, Class 2, style optional; MIL-B-26195, Type II, style and type optional or in a crate conforming to MIL-C-104, type, class, and style optional; as applicable to the weight and size of components procured. Only components of like description shall be packed together within the same container except when specified (see 6.2), components of unlike description in the quantities required for one launcher shall be packed together. The components shall be cushioned, blocked, braced, and anchored, as applicable, within the container to prevent movement. The

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containers shall be closed and strapped in accordance with the specification or appendix to the specification, as applicable. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level C.

5.2.2.1 Procurement of complete launchers. Launchers shall be prepared for shipment as specified in 5.2.1.1.

5.2.2.2 Procurement of launcher components. When only launcher components are procured, they shall be preserved as specified in 5.1, and packed to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Material inspection and receiving report. At time of shipment of each launcher, the Material Inspection and Receiving Report, DD Form 250, shall be annotated in a clear and concise manner as to the level of preservation of the launcher shipped. In addition, it shall be clearly rubber-stamped or stenciled in red ink, all with capital letters 1/8-inch in height, minimum, "VEHICLE AND ENGINE PRESERVED FOR STORAGE. DO NOT CRANK OR OPERATE ENGINE. MOVE BY TOWING OR LIFTING ONLY."

5.4 Marking. Marking of complete launchers and separately procured components shall be marked in accordance with MIL-STD-129. Launchers shall be additionally marked in accordance with the special marking specified in 5.4.1.

5.4.1 Special marking. A warning notice, printed or stenciled on heavy red card stock 12-inches x 12-inches in size in black letters of maximum size shall be prepared as follows:

"VEHICLE AND ENGINE PRESERVED FOR STORAGE. DO NOT CRANK OR OPERATE ENGINE. MOVE BY TOWING OR LIFTING ONLY." This warning notice shall be secured to the steering control bar with tape conforming to PPP-T-60, Type IV.

6. NOTES

6.1 Intended use. The M48A5 and M60A1 bridge launcher-transporter is intended to transport, launch, and retrieve the armored vehicle launched bridge (AVLB) during assault operations in the military theatre of operations.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Bridge launcher-transporter to be furnished, M48A5 or M60A1 or both (see 1.2) or component(s) to be furnished and quantity to be furnished (see 3.1).
(c) Time frame required for submission of the preproduction model M48A5 or M60A1 bridge launcher or component (see 3.2).
(d) Time frame required for submission of the test reports of the hydraulic system components (see 3.2.1).
(e) When initial production testing is required and the quantity of M48A5 or M60A1 bridge launcher required.
(f) When Government furnished property is to be furnished, and the equipment to be furnsihed (see 3.11).
(g) When radio equipment will be furnished by the Government and model of radio set to be furnished (see 3.11).
(h) Level of preservation and packing required (see 5.1 and 5.2).
(i) When unlike components shall be packed together (see 5.2.1.2).
(j) When other than Finish B strapping is required (see 5.2.1.2).

6.3 Preproduction model. Any changes or deviations of production M48A5 or M60A1 bridge launchers or separately procured components from the approved preproduction model during production will be subject to approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish M48A5 or M60A1 bridge launchers or separately procured components conforming to this specification.

6.4 Hydraulic cylinders. The contracting officer should include a requirement that all hydraulic cylinders specified in 3.6.5 furnished for the total quantity of bridge launchers shall be fabricated by the same manufacturer.

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the launcher covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property specified in Table II, and the radio equipment listed either in Table III or Table IV, as specified, when applicable.

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.12 when applicable.

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6.8 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.9 Definitions.

6.9.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.11 Launcher maintenance during tests. The contracting officer should arrange to furnish the technical manual that applies to the launcher undergoing test, M48A5 or M60A1.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-L-52325C(ME) Launcher, Transporter, Bridge: M48A5 or M60A1 Tank Chassis

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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